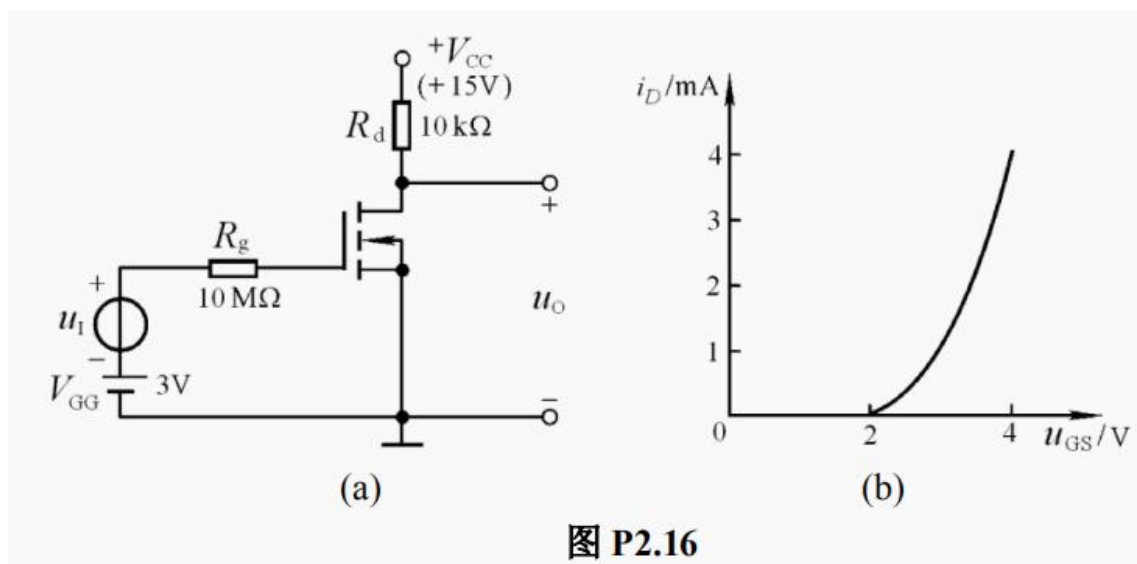


Homework 3

◇ FET, Composite amplifier, Multistage amplifier circuits: 2.16/2.18/3.2

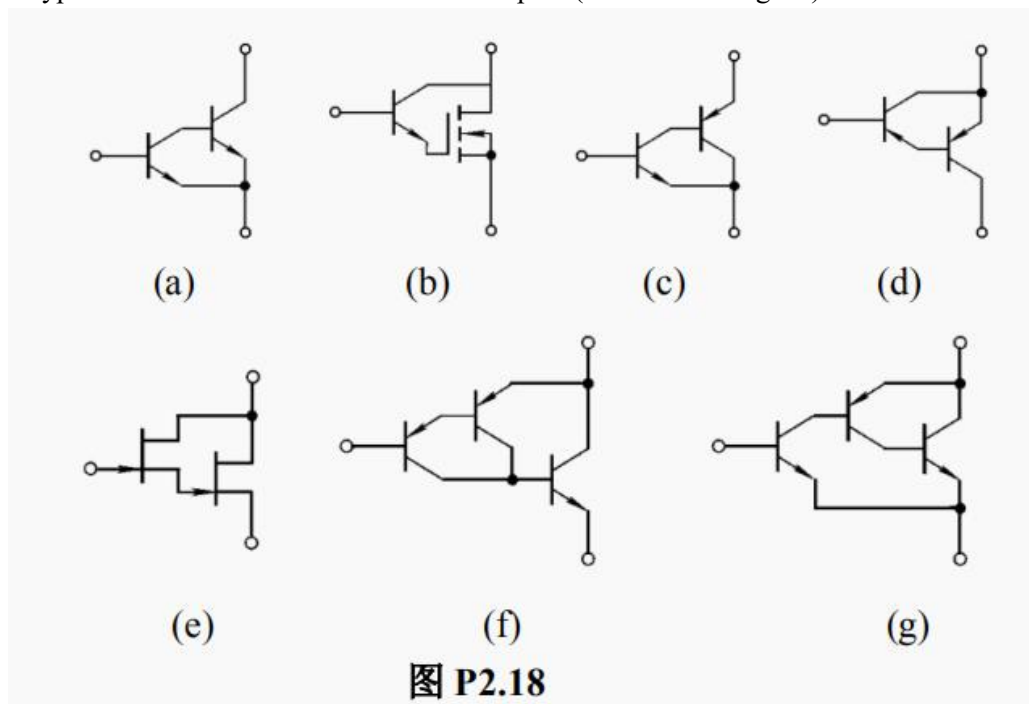
2.16 已知图 P.16(a)所示电路中场效应管的转移特性如图(b)所示。求解电路的静态工作点 Q 点和交流电压增益 A_u 。

2.16 The transfer characteristics of the FET is shown in figure P2.16 (b). Ask: what's the quiescent working point Q and the voltage gain A_u of the circuit.



2.18 图 P2.18 中的哪些接法可以构成复合管?标出它们等效管的类型(如 NPN 型、PNP 型、N 沟道结型.....)及管脚(b、e、c、d、g、s)。

2.18 Which connection in figure P2.18 can make a good composite amplifier? For these good composite amplifier, please show their types (such as NPN type, PNP type, N-type FET, or P-type FET) and mark the three terminals/pins (b、e、c、d、g、s).



3.2 设图 P3.2 所示各电路的静态工作点均合适,分别画出它们的交流等效电路,并写出电压增益 A_u 、输入内阻 R_i 和输出电阻 R_o 的表达式。

3.2 Suppose the quiescent working points of each circuit shown in figure p3.2 are suitable, please draw their AC equivalentent circuits respectively, and write down the expressions of voltage gain A_u , input resistance R_i and output resistance R_o .

